

# Radar Agent

Signals are worth paying for.

An autonomous agent that buys real-time market signals with USDC nanopayments and manages a USDC treasury on Arc — with hard-coded risk discipline from production trading bots.

Encode Club x Circle — Programmable Money Hackathon — Agentic Economy track  
Team: chang chris (solo) · [github.com/CrisChang/radar-agent](https://github.com/CrisChang/radar-agent)

# The problem

Two gaps keep 'AI agent + payments' demos from being real

- **Data has no unit economics for machines.**

Market data is sold in human-sized subscriptions. An agent that needs one signal can't buy one signal.

- **Agents can't be trusted with wallets.**

Most demos are LLM wrappers relaying human commands - no spending limits, no replay safety, no audit trail.

- **Existing chains price out micro-decisions.**

If a query costs cents in gas and seconds in latency, per-signal purchasing can never be economical.

# The solution

A two-sided agentic economy demo on Arc

- **Sell side: paywalled signal API.**  
A real price-alert feed behind x402 - any agent pays per query in USDC via Circle Nanopayments.
- **Buy side: an agent with its own wallet.**  
Decides which signals are worth paying for, buys them, and rebalances a USDC treasury via App Kits.
- **Discipline layer between decision and money.**  
Spending caps, idempotency, circuit breaker, audit trail - every fund-touching action passes through it.

# How it works

## **Seller: Signal API (x402)**

real price-alert feed format  
HTTP 402 -> pay per query in USDC  
Gateway verifies in < 1s  
batch settlement on Arc

**x402 / USDC**

## **Buyer: Radar Agent**

1. pay USDC nanopayment per signal
2. explainable decision rules
3. execute via App Kits (swap/send)
4. discipline layer on every action
5. audit log -> dashboard

Arc Testnet — USDC-denominated gas, sub-second settlement

# Why Arc makes this possible

- **Sub-cent payments need near-zero fees.**  
Paying \$0.001 per signal query only works with USDC-denominated gas and batch-settled nanopayments.
- **Sub-second settlement matches signal latency.**  
A sharp-drop alert is worthless minutes later; verification must be as fast as the signal.
- **Native Circle stack end to end.**  
Nanopayments (x402) + Agent Stack + App Kits + Circle Wallets - no custom payment rails to trust.

# Not a wrapper: the discipline layer

Ported from production trading bots we run daily — already in code

- **Spending caps.**

Separate hard daily budgets for data purchases vs. treasury moves; breach = refuse + alert.

- **Idempotency.**

Deterministic action keys, persisted - a restarted agent can never double-execute.

- **Circuit breaker.**

Consecutive failures or anomalous signal rates halt the agent; only a human can reset.

- **Audit trail.**

Append-only log linking every signal -> decision -> transaction hash, end to end.

# Status & roadmap

- **Done (Checkpoint 2).**

Architecture finalized; discipline layer implemented; signal schema + agent loop skeleton in repo.

- **Week 3.**

x402 nanopayment flow end-to-end on Arc testnet; real signal feed; App Kits treasury execution.

- **Week 4 (final).**

Circuit-breaker demo scenario, dashboard, 3-minute video, MVP deployed on Arc testnet.

- **Beyond the hackathon.**

A marketplace where any radar can sell signals to any agent - unit economics for machine-readable alpha.

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